TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

True Temperature of a Mixture by Radiation Correction

A calorimeter leaks heat while the mixture is still rising to its peak, so the highest temperature seen is not the true mixture temperature. The cooling curve, recorded after the peak, tells how much to add back.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 18 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to find the true temperature of a mixture — the temperature it *would* have reached with no heat exchange with the surroundings — by applying a radiation (cooling) correction obtained graphically. In the method of mixtures, hot water is poured into cold water in a calorimeter. Mixing and stirring take a finite time τ, and throughout that time the warm calorimeter is losing heat by conduction, convection and radiation (the three are lumped together as "radiation loss" in this experiment). The thermometer therefore peaks at θ_2, below the true mixture temperature θ.

By Newton's law of cooling, the rate of loss at any moment is proportional to the excess of the calorimeter's temperature over the room. After the maximum, the temperature falls steadily, and that fall is recorded against time. During the rise the temperature climbed roughly uniformly from θ_1 to θ_2, so its *average* excess — and hence the average rate of cooling during the rise — was about *half* the rate just after the maximum. Therefore:

$$\text{correction } \delta \approx \tfrac{1}{2} \times (\text{fall of temperature during the interval } \tau \text{ after the maximum})$$

$$\text{true mixture temperature } \theta = \theta_2 + \delta$$

The temperatures after the maximum are plotted against time (the **cooling curve**), and the fall in the interval equal to the rise time τ is read straight off the graph. The same graph gives the small correction due to the cold water standing in the calorimeter before mixing, if that wait was long.

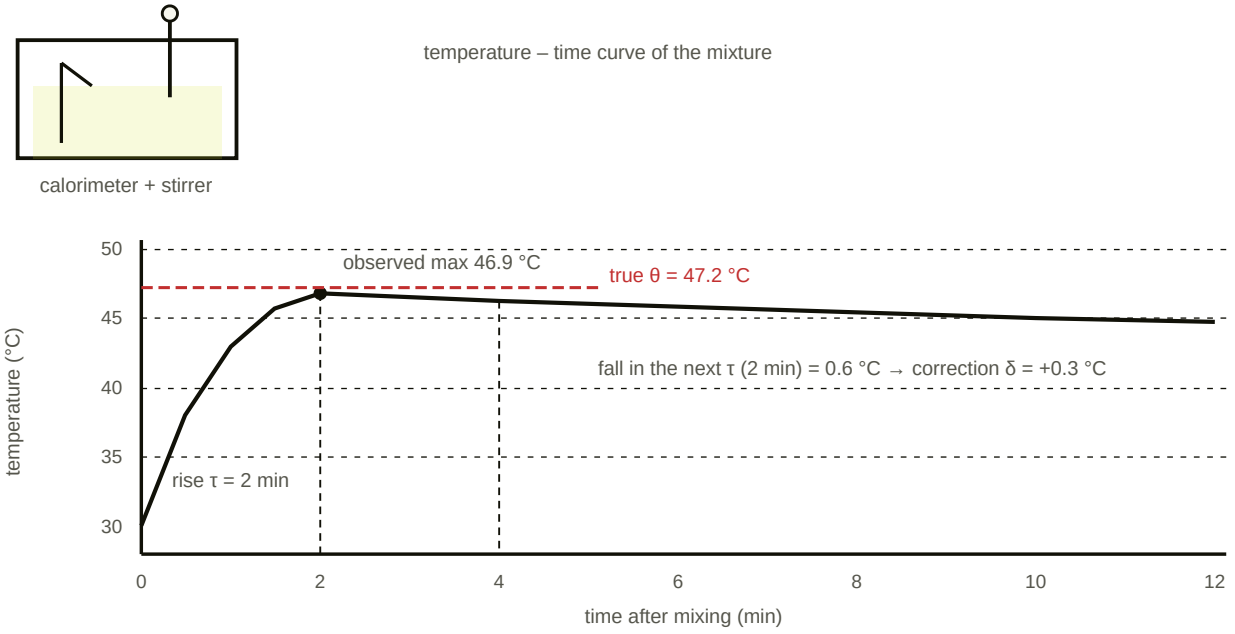


Figure 1 — Inset: the calorimeter. Main figure: the temperature–time curve. The temperature peaks at 46.9 °C after a 2-minute rise, then falls 0.6 °C in the next 2 minutes; half that fall is the radiation correction, giving the true mixture temperature 47.2 °C (dashed line).

2. Apparatus and procedure

- **Apparatus:** calorimeter with stirrer and lagging, two thermometers (0.1 °C), beaker for hot water with heater, stopwatch, balance, graph paper.
- Put a known mass of cold water (200 g) in the calorimeter; record its steady temperature θ_1 (here 30.0 °C) and the water equivalent W of the calorimeter (15 g). Heat the hot water well above the expected mixture temperature.
- Note the hot water's temperature (84.0 °C), pour a known mass (100 g) quickly into the calorimeter, start the stopwatch at the moment of pouring, and stir steadily.
- Read the temperature every 15–30 seconds: follow it up to its maximum θ_2 and note the rise time τ from pouring to the maximum. Then keep reading every 30 seconds for about 10 minutes as it cools.
- Plot temperature against time. From the cooling part, read the fall during the interval τ immediately after the maximum; the correction is half that fall. Add it to θ_2.
- Check: compute the ideal mixture temperature from the heat balance, $(m_h)(84 - \theta) = (m_c + W)(\theta - \theta_1)$, and compare with the corrected value.

3. Worked calculation

Readings: θ_1 = 30.0 °C; observed maximum θ_2 = 46.9 °C, reached τ = 2 min after pouring; in the next 2 minutes the temperature falls from 46.9 °C to 46.3 °C, a fall of 0.6 °C.

$$\delta = \tfrac{1}{2} \times 0.6 = 0.3\ ^\circ\text{C}, \quad \theta = 46.9 + 0.3 = 47.2\ ^\circ\text{C}$$

Heat-balance check: heat lost by 100 g of hot water = heat gained by (200 + 15) g of cold water and calorimeter:

$$\theta = (100 \times 84.0 + 215 \times 30.0) / (100 + 215) = 14850 / 315 \approx 47.1\ ^\circ\text{C}$$

RESULT

True mixture temperature ≈ 47.2 °C, against 47.1 °C from the ideal heat balance — agreement within 0.1 °C. Without the correction the result would have been 46.9 °C, low by 0.3 °C: exactly the sort of systematic error that quietly spoils every specific-heat value computed from an uncorrected mixture temperature.

4. Precautions

- Stir continuously and evenly from pouring to the end of the cooling record; the correction argument assumes the calorimeter is always at one uniform temperature.
- Pour quickly and shield the calorimeter from draughts; minimise τ, because a short rise needs only a small correction.
- Read the thermometer at eye level to 0.1 °C, and time the readings — the whole correction hangs on the temperature–time graph, so sloppy times make a sloppy graph.
- Start with the cold water near room temperature so that before mixing it is neither gaining nor losing heat appreciably; the correction derived here covers the rise period only.

Practice (viva preparation)

1. Why is the correction half the fall in the interval τ after the maximum, rather than the whole fall?
2. A mixture rises for 3 minutes to a maximum of 52.4 °C; in the following 3 minutes it falls to 51.7 °C. Find the true mixture temperature.
3. Why does starting with the cold water a few degrees *below* room temperature reduce the correction?
4. A student stirs only at the moment of reading. What goes wrong with both θ_2 and the correction?
5. State Newton's law of cooling and the assumption about the rise that the graphical method uses.

Answer key

1. During the rise the temperature climbs from θ_1 to θ_2, so its average excess over the cold state — and by Newton's law its average cooling rate — is about half the rate just after the maximum. The heat

lost during the rise therefore equals roughly half the heat lost during the equal interval τ that follows, and temperature corrections follow the heat.

2. Fall = 52.4 − 51.7 = 0.7 °C; δ = 0.35 °C; θ = 52.4 + 0.35 = 52.75 ≈ 52.8 °C.
3. While the mixture is still below room temperature the calorimeter gains heat from the room instead of losing it; the gain during the early part of the rise partly cancels the loss during the later part, shrinking the net correction (the same compensation idea as starting a J experiment below room temperature).
4. Without continuous stirring the liquid is not at one temperature: the thermometer reads its own neighbourhood, θ_2 is unreliable (usually low near the walls, high near the hot inflow), and the cooling after the maximum is not the uniform Newton cooling the method assumes — so both the peak and the correction built on it are wrong.
5. Newton's law: the rate of loss of heat is proportional to the excess of the body's temperature over its surroundings. The method assumes the rise is roughly uniform in time, so the mean excess during the rise is half the final excess — which is what turns the measured post-maximum fall into the half-fall correction.
